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- Introduction
- Bayesian Reasoning
- Bayesian Statistical inference
- Probabilistic Modeling
- Bayesian Computation

Bayesian Statistics and Data Analysis
Lecture 1

Måns Magnusson
Department of Statistics, Uppsala University
Thanks to Aki Vehtari, Aalto University

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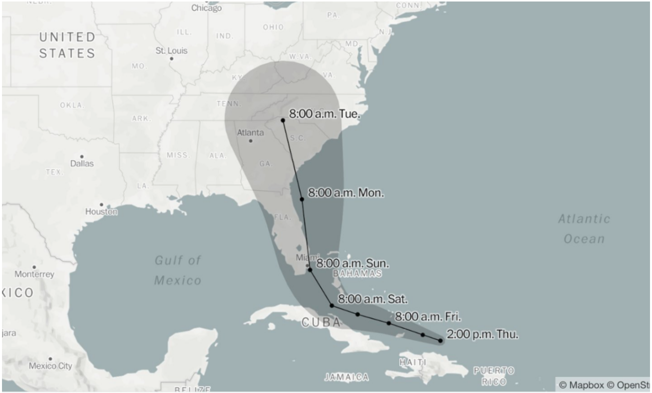
Section 1

Introduction

Decision making in case of uncertainty



Tip! See Matthew Kay's StanCon 2026 keynote:
Systematic uncertainty visualization design

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Quick check: uncertainty in decisions

- Think of one decision from your own life, studies, or research where the outcome is uncertain.

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- What are the possible outcomes?
- What information would change your uncertainty?

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- Pair discussion: what would be the analogue of prior information, data, and a decision?

Bayesian Analysis

- Bayesian probability theory
  - uncertainty is represented with probabilities

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  - probabilities are updated based on new information
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  - considered the problem of *inverse probability*

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  - considered the problem of *inverse probability*
    - significant part of the Bayesian theory
- Bayes did not invent everything, but was first to solve a problem of inverse probability in a special case
- Modern Bayesian theory with rigorous proofs developed in 20th century

Term Bayesian used first time in mid 20th century

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  - the concept of probability was not strictly defined, although it was close to modern Bayesian interpretation
  - at the end of 19th century there was increasing demand for a stricter definition of probability (mathematical and philosophical problem)

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  - term became quickly popular, because alternative descriptions were longer
  - after this Bayesians started to use term "frequentist"

Bayesian core: Uncertainty and probabilistic modeling

- Representing uncertainty with probabilities
- Updating uncertainty (with empirical observations)

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Two types of uncertainty

- Aleatoric uncertainty due to randomness
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Two types of uncertainty

- Aleatoric uncertainty due to randomness
  - observations can't reduce this uncertainty
- Epistemic uncertainty due to lack of knowledge
  - we can reduce this uncertainty with observations
  - two observers may have different epistemic uncertainty

Quick check: aleatoric or epistemic?

- Classify each uncertainty, then compare with your neighbour:
  1. the next roll of a fair die
  2. the unknown fraction of red chips in a bag
  3. tomorrow's temperature in Uppsala

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- Classify each uncertainty, then compare with your neighbour:
  1. the next roll of a fair die
  2. the unknown fraction of red chips in a bag
  3. tomorrow's temperature in Uppsala
- Die roll: mainly aleatoric in this model
- Red-chip fraction: epistemic; sampling chips can reduce it
- Temperature: both; data and forecasts reduce epistemic uncertainty, but weather variation remains

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Section 2

Bayesian Reasoning

Probability in Bayesian Reasoning

- Uncertainty and knowledge are expressed as probability distributions, why?

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Probability in Bayesian Reasoning

- Uncertainty and knowledge are expressed as probability distributions, why?
- Analogy to physical randomness
- Knowledge as coherence of bets (the Dutch book): rationality
  Example: If you say: P(rain) = 0.7 and P(no rain) = 0.4, those add up to 1.1 (too high).
  A bookmaker could then take both bets from you and be sure to profit, because you have made your odds incoherent.
- Accuracy arguments: incoherent credences can be systematically worse estimates of truth
- Conditionalization: update by zeroing possibilities ruled out by evidence and rescaling the rest

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- Accuracy arguments: incoherent credences can be systematically worse estimates of truth
- Conditionalization: update by zeroing possibilities ruled out by evidence and rescaling the rest
- Everyone can have their own 'subjective' uncertainty, e.g. P(rain tomorrow),
  P(Magdalena Andersson will be the next prime minister)
- Frequency arguments can be difficult in some situations: P(other life in the universe)
- See Stanford Encyclopedia of Philosophy: Bayesian Epistemology

Bayesian epistemology

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Bayesian epistemology

- What is epistemology?

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  "the theory of knowledge" (Wikipedia)

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- What is epistemology?
  "the theory of knowledge" (Wikipedia)
- Bayesian epistemology:
  - Studies rational degrees of belief, or credences
  - Two core norms:
    - Probabilism: beliefs at one time should obey the rules of probability
    - Conditionalization: beliefs should be updated by conditioning on new evidence
  - e.g. unlike Popperian approaches, we can talk about P(black swan)
  - Main debates: coherence, priors, idealization, and how evidence should change beliefs
  - Source: Stanford Encyclopedia of Philosophy

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Section 3

Bayesian Statistical inference

Statistical inference

- Draw conclusions about unobserved quantities, based on data

Statistical inference

- Draw conclusions about unobserved quantities, based on data
- Different types of unobserved quantities
  - potentially observed: future observations, potential outcomes,
  - parameters: data-generating process (e.g. regression coefficients)

Notation

- $p(\cdot|\cdot)$: conditional pdf/pmf
- $p(\cdot)$: (marginal) pdf/pmf
- $P(\cdot)$ or $\Pr(\cdot)$: probability, e.g. $P(\theta > 0) = \int_0^\infty p(\theta) d\theta$
- Random variable: $\theta \sim \mathcal{N}(\mu, \sigma^2)$
- pdf/pmf: $p(\theta) = \mathcal{N}(\theta|\mu, \sigma^2)$
- We often suppress known covariates/data x and assumptions h

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Parameters, data, and predictions

- Model parameters: $\theta = (\theta_1, ..., \theta_p)$
- Observed data: $y = (y_1, ..., y_n)$
  - y_i can be a vector and is assumed to come from a data generating process (probabilistic model)
- Potentially observed data: $\tilde{y} = (\tilde{y}_1, ..., \tilde{y}_m)$
- Observed (known) covariates: x

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- Observed (known) covariates: x
- I can be sloppy: e.g. often implicitly condition on x, i.e. $p(\theta|y) = p(\theta|y, x)$

Quick check: model or likelihood?

- Suppose $Y|\theta \sim \text{Bin}(10, \theta)$, so

$$p(y|\theta) = \binom{10}{y} \theta^y (1-\theta)^{10-y}.$$

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- If $\theta = 0.3$ and Y is not observed, what varies?

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- If $y = 7$ is observed and θ varies, what is this function called?

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- If $y = 7$ is observed and θ varies, what is this function called?
- Is $L(\theta|y)$ a probability distribution over θ?

Model and likelihood

- Same expression, different role:

$$p(y|\theta)$$

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- As a function of y with θ fixed: observation model

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$$L(\theta|y) = p(y|\theta)$$

Model and likelihood

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- As a function of y with θ fixed: observation model

$$y|\theta \sim p(\cdot|\theta)$$

- As a function of θ with y fixed: likelihood

$$L(\theta|y) = p(y|\theta)$$

- The likelihood is generally not normalized over θ

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The basic steps of Bayesian inference

1. Setting up a full probability model $p(y|\theta) \cdot p(\theta) = p(y, \theta)$

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The basic steps of Bayesian inference

1. Setting up a full probability model $p(y|\theta) \cdot p(\theta) = p(y, \theta)$
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The basic steps of Bayesian inference

1. Setting up a full probability model $p(y|\theta) \cdot p(\theta) = p(y, \theta)$
2. Conditioning on observed data y to calculate the posterior distribution $p(\theta|y)$
3. Evaluate the model. If not satisfied, go back to 1. Else use the model.

Bayesian Inference I

- Bayesian conclusions about θ or $\tilde{y}$ are made using probabilities, conditional on data y

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- Bayesian conclusions about θ or $\tilde{y}$ are made using probabilities, conditional on data y
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Bayesian Inference I

- Bayesian conclusions about θ or $\tilde{y}$ are made using probabilities, conditional on data y
- We state our uncertainty about θ or $\tilde{y}$ as probability distributions
  - parameters: $p(\theta|y)$
  - potentially observed: $p(\tilde{y}|y)$

Bayesian Inference: Setting up the model

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- Start out with a model: a *joint distribution* for data and parameters:

$$p(y, \theta) = p(y|\theta)p(\theta)$$

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$$p(y, \theta) = p(y|\theta)p(\theta)$$

- $p(y|\theta)$ is our data model, and when conditioned on y, the *likelihood*
- $p(\theta)$ is our prior distribution for our parameters

Bayesian Inference: Computing the posterior

- Conditioning on data y, using *Bayes' theorem*, we can compute the *posterior distribution*

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

where

$$p(y) = \int p(y|\theta)p(\theta)d\theta$$

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- $p(\theta|y)$ summarize our knowledge about θ

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where

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- $p(y)$ is the *marginal likelihood*
- $p(\theta|y)$ summarize our knowledge about θ
- Bayesian statistics obey the *likelihood principle*: data only affects $p(\theta|y)$ through the likelihood $p(y|\theta)$

Predictions

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- How to do inference on an unknown (potential) observable $\tilde{y}$? E.g. a future observation, potential outcome

Predictions

- How to do inference on an unknown (potential) observable $\tilde{y}$? E.g. a future observation, potential outcome
- We use our *data* model and our posterior and *marginalize* over the uncertainty

$$p(\tilde{y}|y) = \int p(\tilde{y}|\theta)p(\theta|y)d\theta$$

- The *posterior predictive distribution*
'An average of conditional predictions over the posterior distribution'

Quick check: updating with chips

- A bag has an unknown fraction θ of red chips.
- Before computing anything, predict what happens to $p(\theta|y)$ after:
    1. one red chip
    2. red, red, red
    3. red, yellow, red, yellow

Quick check: updating with chips

- A bag has an unknown fraction θ of red chips.
- Before computing anything, predict what happens to $p(\theta|y)$ after:
  1. one red chip
  2. red, red, red
  3. red, yellow, red, yellow
- Red observations move the posterior toward larger θ.
- More observations usually make the posterior more concentrated.
- Mixed observations keep the posterior closer to intermediate values.

Example: Updating uncertainty

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- Probability of red $\frac{\#\text{red}}{\#\text{red}+\#\text{yellow}} = \theta$

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- $p(y = \text{red}|\theta) = \theta$ aleatoric uncertainty

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Example: Updating uncertainty

- Probability of red $\frac{\#\text{red}}{\#\text{red}+\#\text{yellow}} = \theta$
- $p(y = \text{red}|\theta) = \theta$ aleatoric uncertainty
- $p(\theta)$ epistemic uncertainty

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Example: Updating uncertainty

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- $p(y = \text{red}|\theta) = \theta$ aleatoric uncertainty
- $p(\theta)$ epistemic uncertainty
- Picking many chips updates our uncertainty about the proportion
- $p(\theta|y_1 = \text{red}, y_2 = \text{yellow}, y_3 = \text{red}, \ldots) =?$

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Example: Updating uncertainty

- Probability of red $\frac{\#\mathrm{red}}{\#\mathrm{red}+\#\mathrm{yellow}} = \theta$
- $p(y = \mathrm{red}|\theta) = \theta$ aleatoric uncertainty
- $p(\theta)$ epistemic uncertainty
- Picking many chips updates our uncertainty about the proportion
- $p(\theta|y_1 = \mathrm{red}, y_2 = \mathrm{yellow}, y_3 = \mathrm{red}, \ldots) =?$
- Bayes rule $p(\theta|y) = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$

Model vs. Likelihood

- Bayes rule

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} \propto p(y|\theta)p(\theta)$$

- Data model: $p(y|\theta)$ as a function of y given fixed θ describes the aleatoric uncertainty

- Likelihood: $p(y|\theta) = L(\theta|y)$ as a function of θ given fixed y provides information about epistemic uncertainty.but is not a probability distribution.

Model vs. Likelihood

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- Likelihood: $p(y|\theta) = L(\theta|y)$ as a function of θ given fixed y provides information about epistemic uncertainty.but is not a probability distribution.

- Bayes rule combines the likelihood with prior uncertainty $p(\theta)$ and transforms them to updated posterior uncertainty $p(\theta|y)$

Example application: Effects on roaches

- Urban pest-management experiment in apartments
- Question: Does treatment reduce the number of captured roaches?

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- Question: Does treatment reduce the number of captured roaches?
- Outcome (y_i): The number of roaches captured in home i
- Covariates (x_i): Treatment, senior home, pre-treatment number of roaches
- Exposure (u_i): Different apartments used traps for different numbers of days
- Source: Gelman and Hill (2007, Ch. 8.3); Vehtari roaches cross-validation demo
- Parameters (θ): Intercept α and coefficients β

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- Source: Gelman and Hill (2007, Ch. 8.3); Vehtari roaches cross-validation demo
- Parameters (θ): Intercept α and coefficients β
- Data model:

$$y_i|\theta, x_i \sim \text{Poisson}(\lambda_i), \quad p(y_i|\theta, x_i) = \frac{\lambda_i^{y_i} \exp(-\lambda_i)}{y_i!}$$

where

$$\lambda_i = u_i \exp(\alpha + x_i^\mathsf{T} \beta)$$

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Example application: Effects on roaches

- Prior:

$$
\begin{aligned}
p(\theta) &= p(\alpha) \prod_{k=1}^{K} p(\beta_k), \\
p(\alpha) &= \mathcal{N}(\alpha | \mu_\alpha, \sigma_\alpha^2), \\
p(\beta_k) &= \mathcal{N}(\beta_k | 0, \sigma_\beta^2)
\end{aligned}
$$

Example application: Effects on roaches

- Prior:

$$p(\theta) = p(\alpha) \prod_{k=1}^{K} p(\beta_k),$$

$$p(\alpha) = \mathcal{N}(\alpha | \mu_\alpha, \sigma_\alpha^2),$$

$$p(\beta_k) = \mathcal{N}(\beta_k | 0, \sigma_\beta^2)$$

- Posterior:

$$p(\alpha, \beta | y, X, u) \propto p(\alpha) \prod_{k=1}^{K} p(\beta_k)$$

$$\prod_{i=1}^{N} \text{Poisson}(y_i | u_i \exp(\alpha + x_i^\mathsf{T} \beta))$$

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Example application: Effects on roaches

- Prior:

$$p(\theta) = p(\alpha) \prod_{k=1}^{K} p(\beta_k),$$

$$p(\alpha) = \mathcal{N}(\alpha|\mu_\alpha, \sigma_\alpha^2),$$

$$p(\beta_k) = \mathcal{N}(\beta_k|0, \sigma_\beta^2)$$

- Posterior:

$$p(\alpha, \beta|y, X, u) \propto p(\alpha) \prod_{k=1}^{K} p(\beta_k)$$

$$\prod_{i=1}^{N} \text{Poisson}(y_i|u_i \exp(\alpha + x_i^{\mathsf{T}}\beta))$$

- Predictive distribution:

$$p(\tilde{y}|\tilde{x}, \tilde{u}, y, X, u) = \int \text{Poisson}(\tilde{y}|\tilde{u} \exp(\alpha + \tilde{x}^{\mathsf{T}}\beta))$$

$$p(\alpha, \beta|y, X, u)\, d\alpha\, d\beta$$

Section 4

Probabilistic Modeling

The art of probabilistic modeling

- Subjectivity: we need to specify both $p(\theta)$ and $p(y|\theta)$
- The art of probabilistic modeling is to describe in a mathematical form (model and prior distributions) what we already know and what we don't know

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- "Easy" part is to use Bayes rule to update the uncertainties
  - computational challenges

The art of probabilistic modeling

- Subjectivity: we need to specify both $p(\theta)$ and $p(y|\theta)$
- The art of probabilistic modeling is to describe in a mathematical form (model and prior distributions) what we already know and what we don't know
- "Easy" part is to use Bayes rule to update the uncertainties
  - computational challenges
- Other parts of the art of probabilistic modeling are, for example,
  - model checking/assessment: is data in conflict with our prior knowledge?
  - model choice: which model should we use?
  - presentation: presenting and visualising the model and the results to the application experts

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Example applications

- Galaxy clusters for cosmology
- Coagulation of blood
- Gene regulation
- Pharmacokinetics and -dynamics
- Decision support
- Effects of nutrition for diabetes
- Evolutionary anthropology
- Clinical trial designs
- Daily demand for gas
- Brain structure trees
- School enrollment
- Sports
- Product demand
- Cocoa bean fermentation
- Marine propulsion power
- Alcohol consumption trends
- Flood probability
- Instantaneous heart rate distributions
- Drug dosing regimens in pediatrics
- Human T stem cell memory cells
- Fairness in university admission policies
- Destruction of bacteria and bacterial spores under heat

Benefits and problems when going Bayesian

- Benefits of Bayesian approach
  - uncertainty and probability theory is the foundation
  - easier interpretation of uncertainty intervals
  - integrate over uncertainties to focus to interesting parts
  - straightforward predictive distributions
  - use relevant prior information
  - straightforward hierarchical models
  - straightforward model checking and evaluation
  - easier to build and evaluate very complex models

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Benefits and problems when going Bayesian

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  - straightforward hierarchical models
  - straightforward model checking and evaluation
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- Complications of Bayesian approach
  - most models do not have nice analytical posteriors
  - we need to *approximate* our posterior
  - can be computationally costly/take long time to estimate complex models

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Section 5

Bayesian Computation

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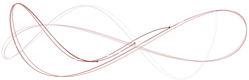
Computation

We want to compute expectations with respect to a posterior distribution $p(\theta|y)$

$$\mathrm{E}_{\theta|y}\left[g(\theta)\right] = \int p(\theta|y)g(\theta)d\theta$$

- Analytic
  - only for very simple models
- Monte Carlo, Markov chain Monte Carlo
  - generic, high-quality
  - computationally expensive
- Distributional approximations
  - e.g. Laplace, variational inference
  - less generic, but can be much faster with sufficient accuracy

Probabilistic programming



Enables agile workflow for developing probabilistic models

language – automated inference – diagnostics

mc-stan.org

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Binomial model for treatment/control comparison

- Two groups of patients: treatment and control
  - Binary outcome (1 success, 0 failure)
  - Is the treatment useful?
- Model:
  Likelihood

$$y_0|\theta_0, N_0 \sim \text{Bin}(N_0, \theta_0)$$
$$y_1|\theta_1, N_1 \sim \text{Bin}(N_1, \theta_1)$$

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  Goal (odds ratio), e.g. $P(OR > 1)$

$$OR = \frac{\frac{\theta_1}{1-\theta_1}}{\frac{\theta_0}{1-\theta_0}}$$

Binomial model for treatment/control comparison

```
data {
  int<lower=0> N1;
  int<lower=0> y1;
  int<lower=0> N0;
  int<lower=0> y0;
}
parameters {
  real<lower=0,upper=1> theta1;
  real<lower=0,upper=1> theta0;
}
model {
  theta1 ~ beta(1,1);
  theta0 ~ beta(1,1);
  y1 ~ binomial(N1, theta1);
  y0 ~ binomial(N0, theta0);
}
generated quantities {
  real oddsratio;
  oddsratio = (theta1/(1-theta1))/(theta0/(1-theta0));
}
```

Binomial model for treatment/control comparison

RStanARM

```
fit_bin2 <- stan_glm(cbind(y, N - y) ~ grp2,
                    family = binomial(), data = d_bin2)
```

Modeling nature

- Drop a ball from different heights and measure time (data)

Modeling nature

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  - air resistance, air pressure, shape and surface structure of the ball
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- Taking into account the accuracy of the measurements, how accurate model is needed?

Modeling nature

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- Drop a ball from different heights and measure time (data)
  - Newton
  - air resistance, air pressure, shape and surface structure of the ball
  - relativity
- Taking into account the accuracy of the measurements, how accurate model is needed?
  - often simple models are adequate and useful
  - *All models are wrong, but some of them are useful*, George P. Box

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Recap: Uncertainty and probabilistic modeling

- Two types of uncertainty: aleatoric and epistemic
- Representing uncertainty with probabilities
- Updating uncertainty using data

Rest of the course

- Basic models which can be used as building blocks for complex ones
- Basic computation of posterior and predictive distributions
- Typical simple scientific data analysis cases
  - e.g. comparison of treatments
- Presentation of the results

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Week 1 reading

- BDA3 Chapter 1
  - 1.1–1.3: steps, notation, Bayes' rule
  - 1.4: discrete examples
  - 1.5: probability as uncertainty
  - 1.6: calibration example
  - 1.8: probability review for assignments
- Focus on the terms in the reading instructions